Bakshree Mishra

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Research Statement

My research focuses on efficient orchestration of LLM inference across multiple heterogeneous edge and cloud devices, addressing performance, scalability, and data-movement bottlenecks at system and architectural levels.

Education

PhD in Computer Science — **2021-Present**
University of Illinois Urbana-Champaign — Gpa: **3.92/4**
Advisor: Prof. Sarita Adve (*sadve@illinois.edu*)
Research Interests: Computer Architecture, Hardware-Software Codesign, Machine Learning & Systems

M.Tech in Computer Science — **2015-2017**
National Institute of Technology, Rourkela — Gpa: **9.69/10**
Advisors: Prof. Bansidhar Majhi (NIT Rourkela), Mr. Tarjinder Singh (Intel)

B.Tech in Computer Science and Engineering — **2010-2014**
College of Engineering and Technology, Bhubaneswar — Gpa: **8.85/10**

Patents

- Boschi, G. and Makkadayil, S. and Manjunath, R. and **Mishra, B.** and Campinoti, A. "Register Fault Detector." U.S. Patent US12,373,295 B2, issued 2025.
- Singh, T. and SR, S. and Sumiran, R. and **Mishra, B.** and Makkadayil, S. and Thyagarajan, V. and Baireddy, V. "Graph Reordering and Tiling Techniques." U.S. Patent Application 17/533,976, filed 2021.
- Makkadayil, S. and Paul, S. and Saifee, S. and **Mishra, B.** and Thyagarajan, V. and Velayudha, M. and Khellah, M. and Udofia, A. "Parallel Pruning and Batch Sorting for Similarity Search Accelerators." U.S. Patent Application 17/358,495, filed 2021.

Publications

- **Mishra, B.**, Hou, E., Pattisapu, S., Adve, S., *Title Redacted for Double Blind Review*, **submitted to ASPLOS'27**.
- **Mishra, B.**, Alsop, J., Boyer, M., Choi, J., Adve, S., *Title Redacted for Double-Blind Review*, **Submitted to MICRO'26**.
- **Mishra, B.**, *Minstrel: Application-Aware SLM Inference Optimization on Edge Devices*, **MLArchSys'26**.
- Suresh, V. and **Mishra, B.** and Jing, Y. and Zhu, Z. and Jin, N. and Block, C. and Mantovani, P. and Giri, D. and Zuckerman, J. and Carloni, L. and Adve, S. *Mozart: Taming Taxes and Composing Accelerators with Shared-Memory* In Proceedings of the 2024 International Conference on Parallel Architectures and Compilation Techniques (PACT '24). [Paper]
- **Mishra, B.** and Chakraborty, D. and Makkadayil, S. and Patil, S. D. and Nallani, B. *Hardware Acceleration of Computer Vision and Deep Learning Algorithms on the Edge using OpenCL*. EAI Endorsed Transactions on Cloud Systems, 2019. [Paper]

Work Experience

Graduate Research Assistant — *University of Illinois Urbana-Champaign* — May 2022 – Present
Research on heterogeneous and disaggregated accelerator systems for efficient LLM inference; developed analytical and empirical performance, communication, and energy models; built systems for real-time language model inference over heterogeneous edge systems.

Research Associate — *AMD, Bellevue* — May 2025 – Nov 2025
Graduate (PhD) Intern — *AMD, Bellevue* — May 2024 – Aug 2024
Analyzed NoC and data-movement bottlenecks in 3D-stacked GPU architectures across a variety of workloads. Developed analytical models for LLM inference on these architectures under different orchestration strategies.

ML and IP Design Engineer — *Intel Corporation, Bangalore* — Jun 2017 – Aug 2021
Design and FPGA validation for ML accelerators targeted for production Intel platforms. Developed multiple demos that were demonstrated to leadership, presented at Intel DTTC'19, co-authored three patent applications, and won several Departmental and Divisional recognition awards.
Previously: Graduate Technical Intern, Intel (May 2016 – May 2017).

Assistant System Engineer — *Tata Consultancy Services, Bhubaneswar* — Jun 2014 – Jul 2015
Summer Intern — *Tata Consultancy Services, Bhubaneswar* — Jun 2013 – Aug 2013
Worked as a full-stack software engineer for an e-governance project, where I owned and shipped two modules.

Teaching Experience

Graduate Teaching Assistant — *University of Illinois Urbana-Champaign* — Aug 2021 – May 2022

CS 233, Computer Architecture: *Fall'21*	Was TA for the regular and honors sections, oversaw discussions and hands-on labs, performed office hours and grading.
CS 225, Data Structures: *Spring 22*	Led three well-received lecture-discussion sections and office hours per week. Was recognized among Teachers Ranked as Excellent (Spring 2022).

Projects

Minstrel: Orchestrating Language Model Inference across edge devices *2024–Present*
• Developed an analytical–empirical model for predicting LLM inference latency in a multi-device edge setup.
• Analyzed inference partitioning, KV-cache reuse, and migration trade-offs in multi-device edge deployments.
• Early results published as a workshop paper; full manuscript under peer review.

Investigating NoC Bottlenecks on 3D Stacked GPU Architectures *2025*
• Identified NoC power-density and scalability bottlenecks limiting bandwidth utilization in 3D-stacked GPU.
• Developed an analytical model capturing LLM inference latency and communication behavior under alternative interconnect organizations.
• Proposed communication algorithms inspired by spatial accelerators and performed trade-off analyses.

Mozart: Composable Acceleration with Disaggregated Accelerator Systems *(PACT 2024)* *2021–2024*
• Designed a lightweight accelerator synchronization interface (ASI) reducing invocation overhead in disaggregated accelerator pipelines.
• Evaluated chained and pipelined accelerator compositions across diverse workloads on FPGA.
• Demonstrated performance trade-offs with monolithic and scalability of disaggregated acceleration.

Selected Research Projects at Intel *2016–2021*
• Designed and deployed OpenCL-based FPGA accelerators for real-time computer vision workloads, achieving up to 5× throughput improvement; work was presented at various internal demos, Intel DTTC, IEEE WinTechCon, and published in *EAI Endorsed Transactions on Cloud Systems*, 2019.
• Developed functional safety IP including RTL design, waveform analysis, and verification; granted U.S. patent.

Select Awards and Honors

• Selected for CRA-WP Grad Cohort program (one in 300) — 2022
• Among Teachers Ranked as Excellent for CS225 — Spring 2022
• Co-authored 3 accepted internal papers, presented a demo at Intel DTTC, Portland, OR — 2019-2021
• Multiple Intel Divisional and Departmental Recognition Awards — 2017-2020
• Ranked 2^{nd} among all Masters (~110) students in CS Department at NIT Rourkela — 2017
• CET Merit Scholarship (Undergrad scholarship 2010-2014) — 2010
• National Talent Search Examination Scholar and CBSE X^{th} board rank-holder in State — 2008

Technical Skills

• **Modeling & Analysis:** Analysis of ML workloads over CPU, GPU, and heterogeneous accelerators
• **Prototyping:** Implement analytical prototypes as well as deployable systems using Python and C/C++
• **Hardware Design:** Understanding as well as developing specification documents; RTL, SystemC, and OpenCL/HLS-based hardware design, waveform analysis, FPGA deployment, and end-to-end system evaluation
• **Benchmarking:** Power profiling, DVFS analysis, and application benchmarking on heterogeneous systems

Service, Mentoring, & Professional Development

• CRA Graduate Mentor for the UR2PhD Program [Certificate] — Fall'25 - present
• Planning committee member of GradWCS, hosting monthly Faculty/Student luncheons — 2023 - present
• Co-started and run weekly coffee meet-ups for women in comp. arch in CS and ECE — 2022 - present
• Co-chair of the Systems and Architecture session for the 19^{th} CSL Student Conference — 2024
• Mentored four MS interns for their graduate internship and thesis research — 2017 - 2021
• Named one of Top 50 Volunteers in Intel India for community service initiatives — 2020
• Received Intel Seed Grant to renovate the nurses' dining hall at Karunashraya Hospice — 2019